Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Dual Vector Gene Delivery for Large-Gene Rare Genetic Diseases			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio・Medical		Bio・Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Gene Pharmaceuticals	Medical Bio	Pharmaceutical Bio-based Technologies and Systems
Concept of Proposed Technology	Thousands of rare genetic diseases are caused by **loss-of-function mutations** in genes that cannot be addressed using conventional therapeutic modalities. Antibody- and small-molecule approaches require druggable protein targets, while RNA-based therapeutics depend on the presence and regulation of endogenous genes, rendering these approaches unsuitable for a substantial fraction of monogenic disorders. Gene therapy represents a transformative solution for such diseases; however, current clinical-stage and approved gene therapies rely predominantly on **single adeno-associated virus (AAV) vectors**, which are constrained by a strict packaging limit of approximately 4.7 kb[1]. As a result, many clinically relevant genes remain inaccessible to existing AAV-based approaches. Genomic analyses indicate that **hundreds to over one thousand rare monogenic diseases** are caused by genes whose coding sequences exceed single-vector capacity[2]. These disorders affect critical organs such as the central nervous system, skeletal muscle, and cardiac muscle and frequently present early in life with no disease-modifying treatment options.			

	To overcome this fundamental limitation, **multi-vector gene delivery strategies**, particularly **dual-vector or split-gene systems**, have emerged as a promising and broadly applicable solution[3,4]. By dividing large therapeutic genes into multiple components that are delivered separately and reconstituted in vivo, dual-vector systems enable functional expression of genes otherwise beyond reach. Importantly, payload limitation is a **shared structural constraint** across many rare genetic diseases. Advances in dual-vector engineering, reconstitution efficiency, and expression control therefore have the potential to impact multiple therapeutic areas simultaneously. The proposed program aims to establish a **scalable and reusable development framework for large-gene gene therapies**, combining advanced dual-vector engineering with **structured human-relevant translational validation**. This approach enables efficient adaptation across biologically related rare disease indications while maintaining a clear path toward clinical and commercial feasibility.
Rationale for Joint R&D Project	While Korean SMEs often possess strong capabilities in construct engineering and rapid proof-of-concept execution, translation of large-gene gene delivery technologies into globally competitive therapeutic programs remains challenging. First, many large-gene rare diseases—particularly those affecting the central nervous system—exhibit **low translational predictability in conventional animal models[5]**. Standard mouse models frequently fail to recapitulate human-specific gene regulation, expression fidelity, and functional rescue, creating substantial risk during clinical advancement. Second, the **mechanistic biology of split-gene reconstitution** remains incompletely understood. Factors such as cell-type–specific expression control, driven by promoter and capsid selection, and long-term expression dynamics require advanced experimental systems and analytical expertise typically unavailable within SME environments[6,7]. Third, global commercialization requires **early integration of first-in-human–oriented design principles**, including dosing logic, tissue specificity trade-offs, and long-term safety interpretation. These considerations are most effectively addressed through collaboration with overseas academic medical centers equipped with human-relevant disease models and translational research infrastructure. International joint R&D is therefore essential to reduce translational risk, strengthen mechanistic understanding, and support globally competitive development of large-gene therapies.
Key Words	1. Large-gene gene therapy 2. Dual-vector gene delivery 3. Rare monogenic diseases 4. Human-relevant disease models 5. Translational validation

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2)]	**Start of the Project**	Level (TRL 2)	**End of the Project**	Level (TRL 5)

R&D Objectives	• Establish robust dual-vector systems capable of delivering large therapeutic genes exceeding single-vector capacity • Elucidate mechanistic principles governing split-gene reconstitution and functional restoration • Optimize cell-type–specific expression through promoter and vector design • Validate platform performance using clinically predictive human-relevant disease models • Integrate human-translational design principles early in development to support future clinical advancement
R&D Target Specifications	• Reliable in vivo reconstitution of large genes exceeding single-vector capacity (e.g., >30% reconstitution compared to the normal expression level in target tissues) • Demonstrated functional rescue in human-relevant cellular models (e.g., >30% of functional rescue compared to the wild-type control in a disease model) • Mechanistic understanding sufficient to guide rational vector and promoter optimization • Translational readiness aligned with early clinical development requirements (e.g., submission of an initial regulatory meeting, such as FDA INTERACT or Pre-IND meeting, or its equivalent in other countries)
R&D Plan	**Year 1: Platform Establishment and Feasibility Validation** • Dual-vector construct design and optimization • Establishment of human-relevant cellular and organoid models • In vitro proof-of-concept studies **Year 2: Mechanistic Studies and Translational Refinement** • Analysis of split-gene reconstitution mechanisms • Cell-type–specific expression optimization • In vivo proof-of-concept studies • Comparative evaluation between animal and human-relevant models **Year 3: Translational Integration and Generalization** • Integration of first-in-human–oriented design considerations • Evaluation of long-term expression and safety-related parameters • Assessment of cross-indication applicability and platform generalizability

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Development and application of patient-derived iPSC, organoid, and advanced neuronal models • Execution of in vitro proof-of-concept studies and mechanistic analysis of large-gene reconstitution using a human-relevant model developed • Guidance on human dosing logic and tissue specificity considerations • Interpretation of long-term expression and safety-related data

Expected R&R of Korean SME Partner	• Engineering and optimization of dual-vector constructs • Execution of in vivo proof-of-concept studies • Integration of mechanistic and translational insights into platform development • Platform expansion and planning for global commercialization

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project is expected to establish a **globally competitive large-gene gene therapy development framework** supported by human-relevant translational validation. The resulting platform will significantly reduce translational risk, shorten development timelines, and enable Korean SMEs to advance multiple large-gene therapeutic programs toward global clinical development. From a broader perspective, the project contributes to strengthening national competitiveness in advanced gene therapy technologies, expanding opportunities for international collaboration, and creating new pathways for global commercialization in rare disease therapeutics.

6. References

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